Phase 1 — Descriptive

- Views are heavily right-skewed (mean 4.28M vs median 3.47M, skew = positive) — mention this justifies the log-transformation used in the model
- The correlation heatmap will show `likes` strongly correlates with `views`

Phase 2 — Inferential

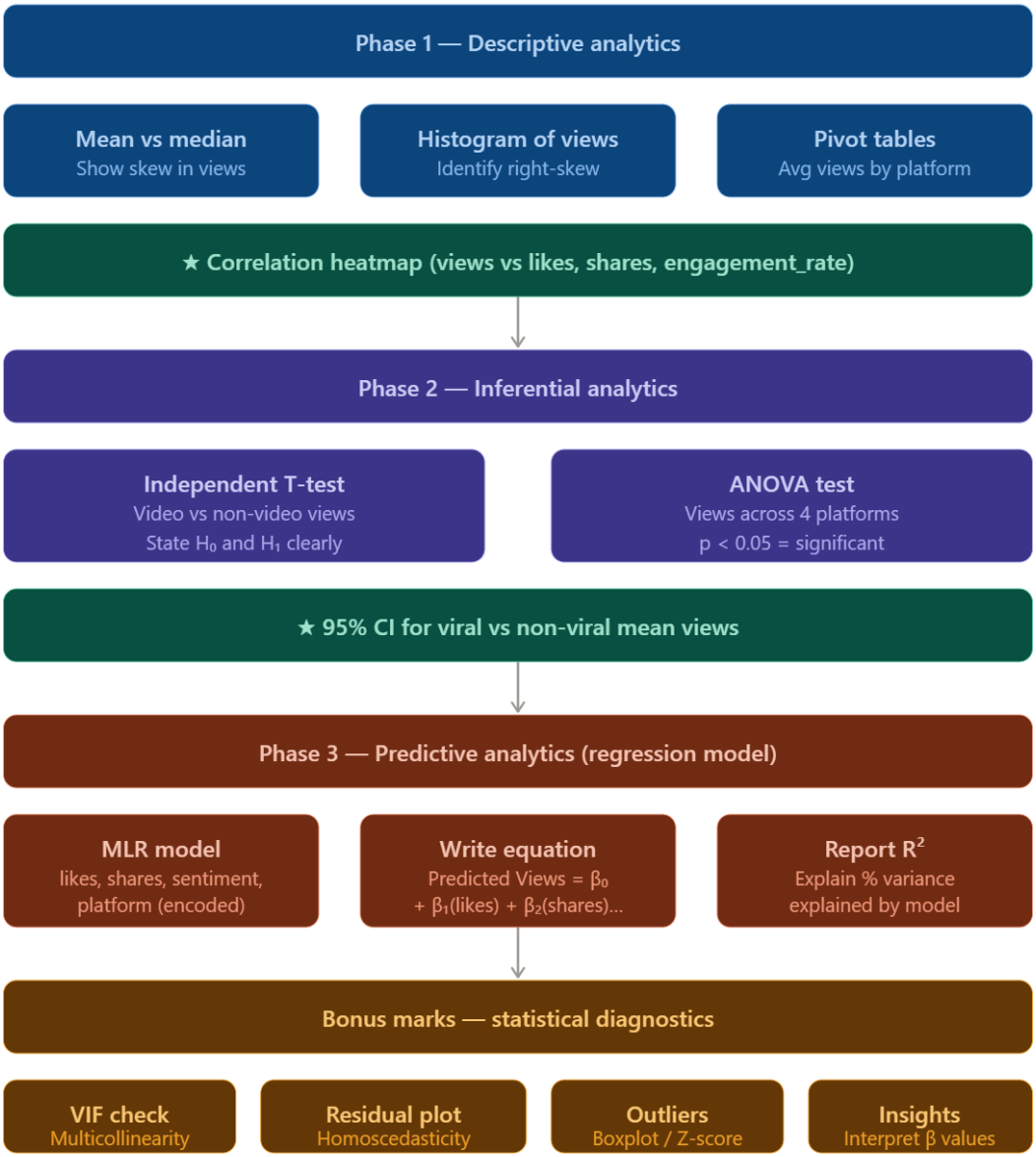
- T-test (video vs non-video): p = 0.058 — just above 0.05, so you *fail to reject* H_0. Write this as: "there is no statistically significant difference at α=0.05, though the result approaches significance"
- ANOVA (platforms): p = 0.93 — platforms perform similarly. Write: "platform alone does not significantly predict views"
- CI comparison: Viral posts average **5.69M views** vs non-viral **1.01M** — the confidence intervals don't overlap at all, which is a powerful finding to highlight

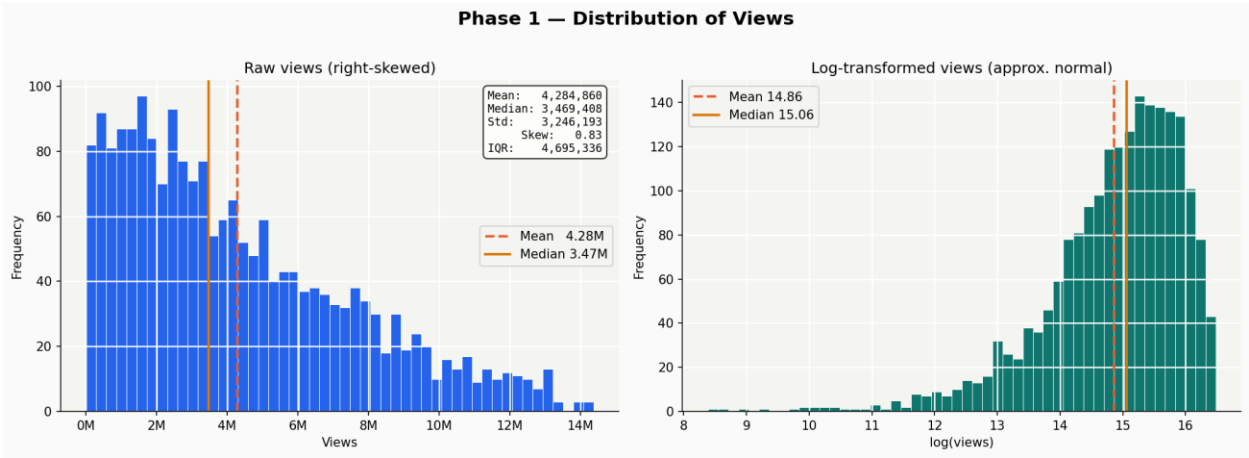
Phase 3 — Regression

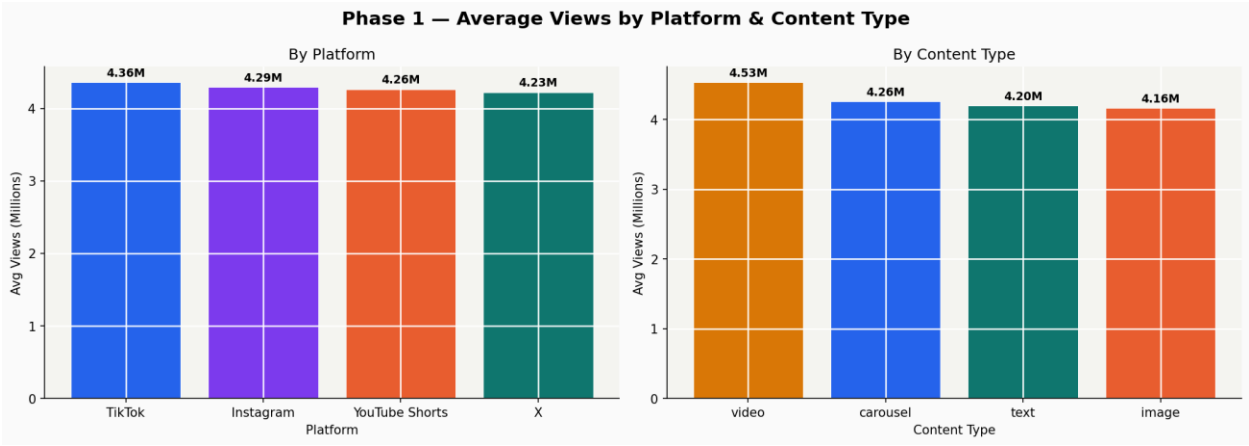
- R^2 = **0.731** → your model explains 73.1% of variance in log(views) — excellent for social media data
- `likes` (p < 0.001) and `is_viral` (p < 0.001) are the only statistically significant predictors
- Write the equation as: *Predicted log(views) = 13.219 + 0.0000034 × likes + 1.159 × is_viral + …*
- Actionable insight: "A post being viral adds approximately e^1.159 ≈ **3.19× more views**, holding all else constant"

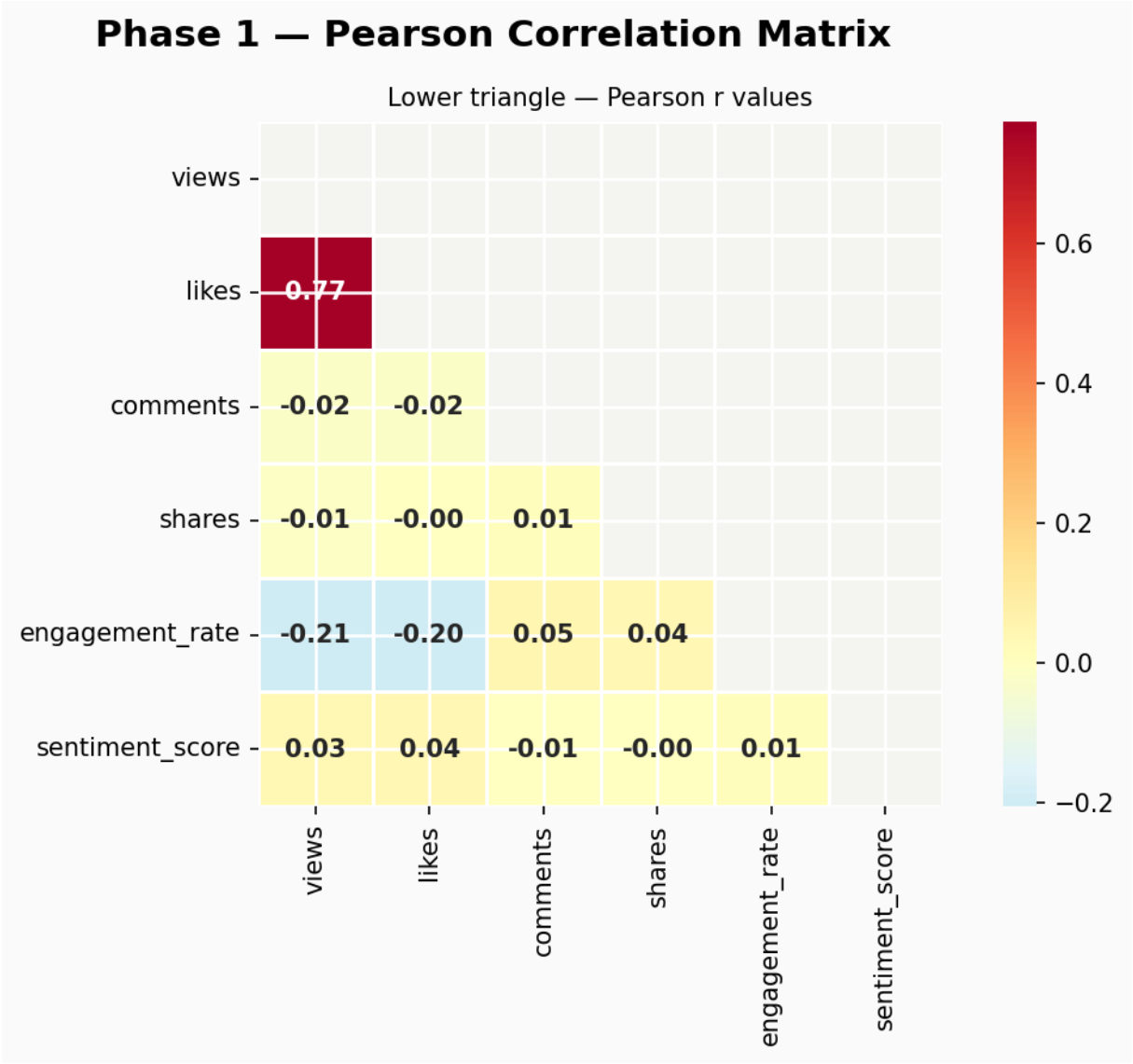
Bonus Diagnostics

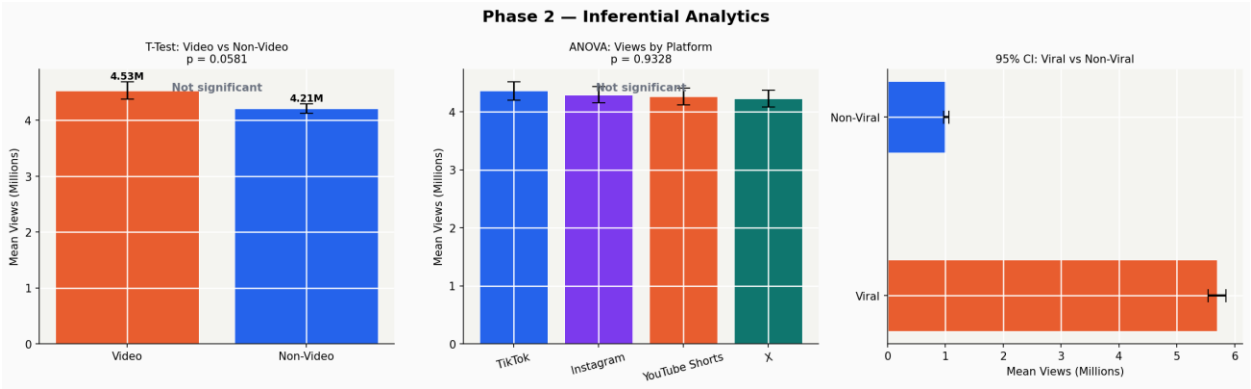
- VIF values are all below 2 — no multicollinearity at all. Write: "All VIF < 5, confirming no problematic multicollinearity"
- Only 3 outliers (0.1%) by Z-score — justify keeping them as genuine viral behavior
- The residual plot and Q-Q plot complete your diagnostic section

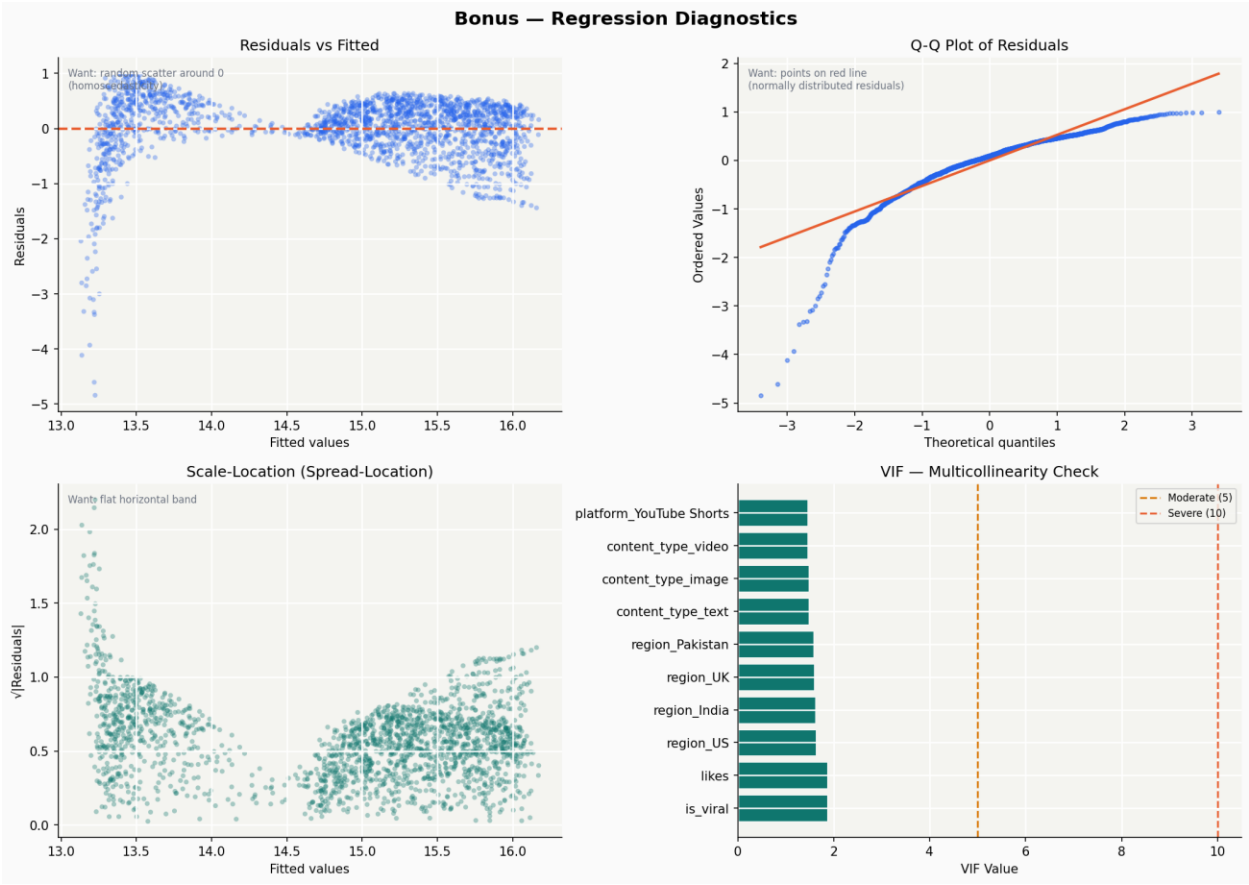

Phase 1 — Descriptive analytics
Mean vs median
Show skew in views
Histogram of views
Identify right-skew
Pivot tables
Avg views by platform
★ Correlation heatmap (views vs likes, shares, engagement_rate)
Phase 2 — Inferential analytics
Independent T-test
Video vs non-video views
State H_0 and H_1 clearly
ANOVA test
Views across 4 platforms
$p < 0.05$ = significant
★ 95% CI for viral vs non-viral mean views
Phase 3 — Predictive analytics (regression model)
MLR model
likes, shares, sentiment, platform (encoded)
Write equation
Predicted Views = $\beta_0 + \beta_1$(likes) + β_2(shares)...
Report R^2
Explain % variance explained by model
Bonus marks — statistical diagnostics
VIF check
Multicollinearity
Residual plot
Homoscedasticity
Outliers
Boxplot / Z-score
Insights
Interpret β values

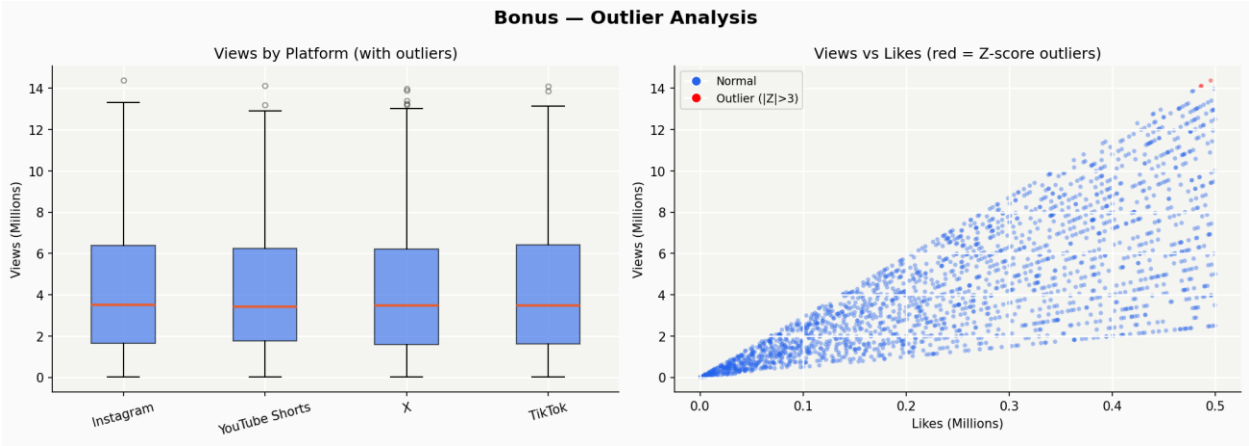

Phase 1 — Distribution of Views
Raw views (right-skewed)
Mean: 4,284,860
Median: 3,469,408
Std: 3,246,193
Skew: 0.83
IQR: 4,695,336
Mean 4.28M
Median 3.47M
Frequency
Views
0M
2M
4M
6M
8M
10M
12M
14M
Log-transformed views (approx. normal)
Mean 14.86
Median 15.06
Frequency
log(views)


Phase 1 — Average Views by Platform & Content Type
By Platform
4.36M
4.29M
4.26M
4.23M
Avg Views (Millions)
TikTok
Instagram
YouTube Shorts
X
Platform
By Content Type
4.53M
4.26M
4.20M
4.16M
Avg Views (Millions)
video
carousel
text
image
Content Type


Phase 1 — Pearson Correlation Matrix
Lower triangle — Pearson r values
views
likes
comments
shares
engagement_rate
sentiment_score
0.77
-0.02 -0.02
-0.01 -0.00 0.01
-0.21 -0.20 0.05 0.04
0.03 0.04 -0.01 -0.00 0.01
0.6
0.4
0.2
0.0
−0.2


Phase 2 — Inferential Analytics
T-Test: Video vs Non-Video
p = 0.0581
4.53M
4.21M
Not significant
Mean Views (Millions)
Video
Non-Video
ANOVA: Views by Platform
p = 0.9328
Not significant
TikTok
Instagram
YouTube Shorts
X
95% CI: Viral vs Non-Viral
Non-Viral
Viral
Mean Views (Millions)


Bonus — Regression Diagnostics
Residuals vs Fitted
Want: random scatter around 0
(homoscedasticity)
Residuals
Fitted values
Q-Q Plot of Residuals
Want: points on red line
(normally distributed residuals)
Ordered Values
Theoretical quantiles
Scale-Location (Spread-Location)
Want: flat horizontal band
√|Residuals|
Fitted values
VIF — Multicollinearity Check
Moderate (5)
Severe (10)
platform_YouTube Shorts
content_type_video
content_type_image
content_type_text
region_Pakistan
region_UK
region_India
region_US
likes
is_viral
VIF Value


Bonus — Outlier Analysis
Views by Platform (with outliers)
Views (Millions)
Instagram
YouTube Shorts
X
TikTok
Views vs Likes (red = Z-score outliers)
Normal
Outlier (|Z|>3)
Views (Millions)
Likes (Millions)

Step 1 — Feature Engineering (10 new features added)

Time features like `hour`, `day_of_week`, `is_weekend` capture when posts are published. Hashtag features (`hashtag_count`, `has_viral_tag`, `has_ai_tag`) add content signal. Note that `like_ratio`, `comment_ratio`, `share_ratio` were created but then **excluded from training** — using them would be data leakage since they're derived from `views` (the target).

Step 2 — Preprocessing

80/20 train/test split, stratified on `is_viral` so both sets have proportional viral/non-viral posts. Categorical columns encoded as dummies. Linear models use `StandardScaler`; tree models don't need it.

Step 3 — Model Comparison Results

Model	R^2	CV R^2
Linear Regression	0.787	0.705
Ridge Regression	0.787	0.705
Random Forest ★	**0.837**	**0.857**
XGBoost	0.824	0.838

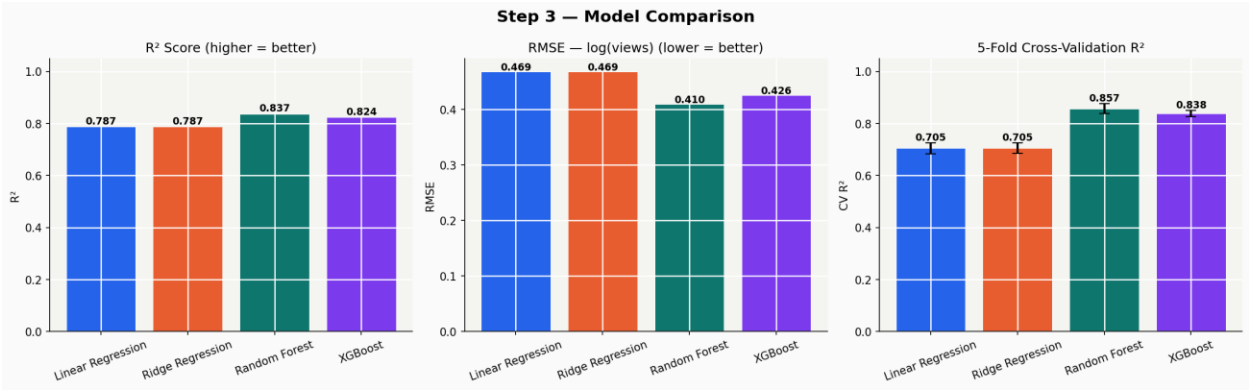
Random Forest wins — 83.7% of variance explained, confirmed by 5-fold cross-validation.

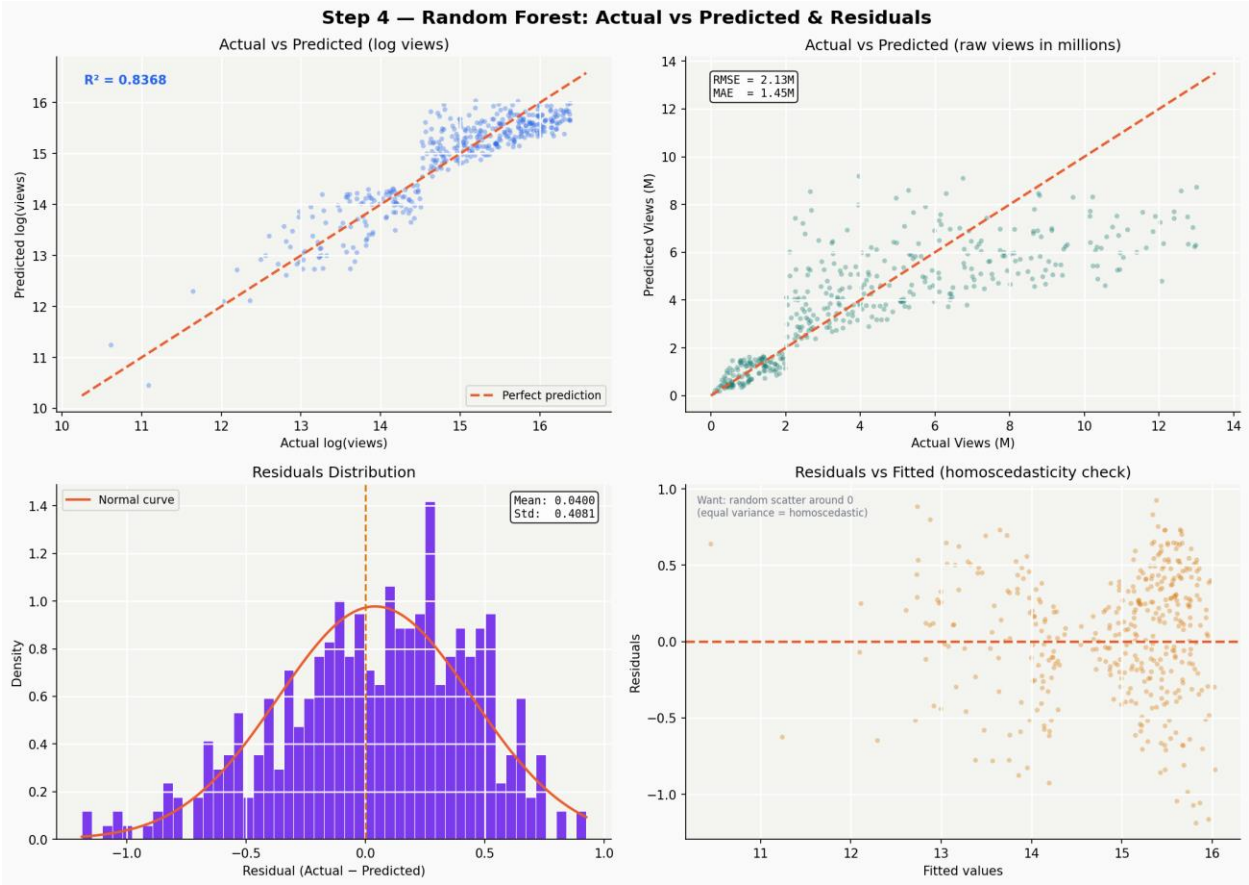
Step 5 — Feature Importance (key finding for your report)

`is_viral` (61.8%) and `likes` (28.7%) dominate — together they explain almost 90% of the model's predictive power. Everything else contributes marginally. Write this as: *"Whether a post goes viral is the single strongest predictor of views, followed by like count."*

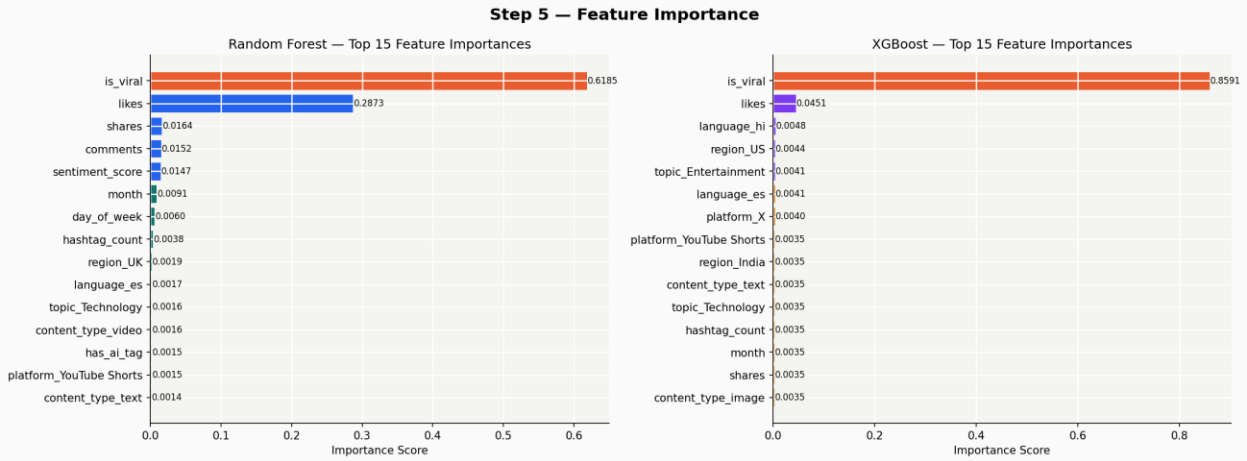
Step 7 — Learning Curve

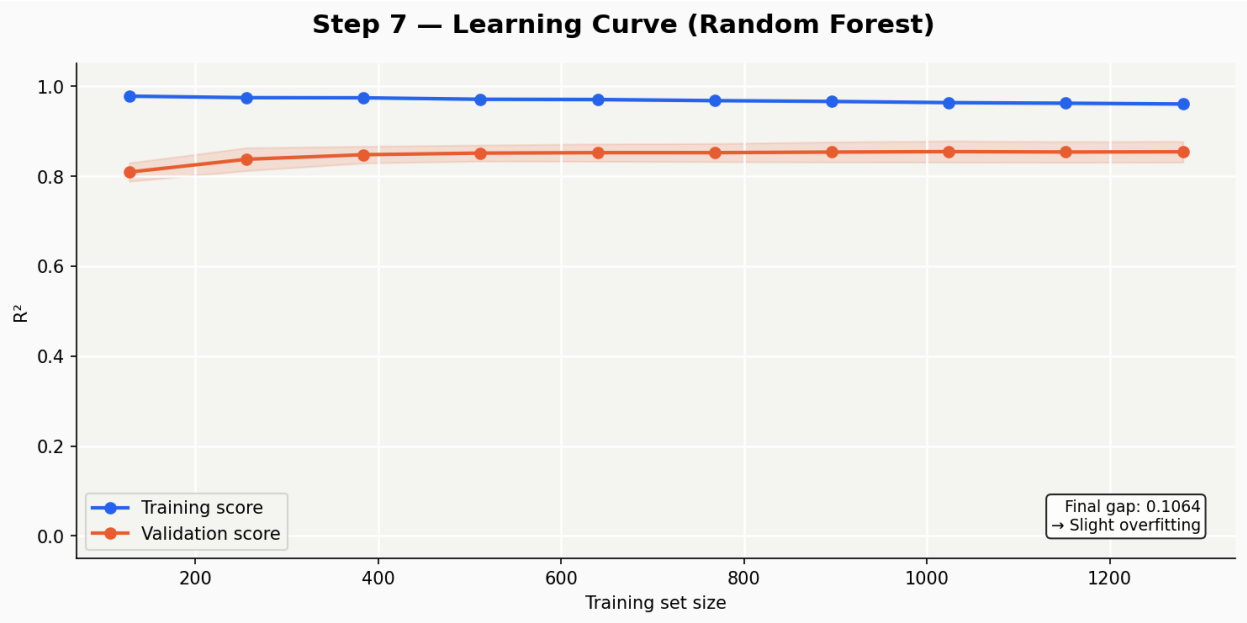
Training R^2 = 0.96 vs Validation R^2 = 0.85 — gap of ~0.10, which means slight overfitting. This is normal for Random Forest and worth mentioning in your report as a limitation, along with the suggestion that more data would reduce the gap.


Step 3 — Model Comparison
R² Score (higher = better)
RMSE — log(views) (lower = better)
5-Fold Cross-Validation R²
R²
RMSE
CV R²
0.787
0.787
0.837
0.824
0.469
0.469
0.410
0.426
0.705
0.705
0.857
0.838
Linear Regression
Ridge Regression
Random Forest
XGBoost


Step 4 — Random Forest: Actual vs Predicted & Residuals
Actual vs Predicted (log views)
R² = 0.8368
Predicted log(views)
Actual log(views)
Perfect prediction
Actual vs Predicted (raw views in millions)
RMSE = 2.13M
MAE = 1.45M
Predicted Views (M)
Actual Views (M)
Residuals Distribution
Normal curve
Mean: 0.0400
Std: 0.4081
Density
Residual (Actual − Predicted)
Residuals vs Fitted (homoscedasticity check)
Want: random scatter around 0
(equal variance = homoscedastic)
Residuals
Fitted values

Public


Step 5 — Feature Importance
Random Forest — Top 15 Feature Importances
is_viral 0.6185
likes 0.2873
shares 0.0164
comments 0.0152
sentiment_score 0.0147
month 0.0091
day_of_week 0.0060
hashtag_count 0.0038
region_UK 0.0019
language_es 0.0017
topic_Technology 0.0016
content_type_video 0.0016
has_ai_tag 0.0015
platform_YouTube Shorts 0.0015
content_type_text 0.0014
0.0 0.1 0.2 0.3 0.4 0.5 0.6
Importance Score
XGBoost — Top 15 Feature Importances
is_viral 0.8591
likes 0.0451
language_hi 0.0048
region_US 0.0044
topic_Entertainment 0.0041
language_es 0.0041
platform_X 0.0040
platform_YouTube Shorts 0.0035
region_India 0.0035
content_type_text 0.0035
topic_Technology 0.0035
hashtag_count 0.0035
month 0.0035
shares 0.0035
content_type_image 0.0035
0.0 0.2 0.4 0.6 0.8
Importance Score


Step 7 — Learning Curve (Random Forest)
R²
1.0
0.8
0.6
0.4
0.2
0.0
Training score
Validation score
Final gap: 0.1064
→ Slight overfitting
200 400 600 800 1000 1200
Training set size